

General Chemistry

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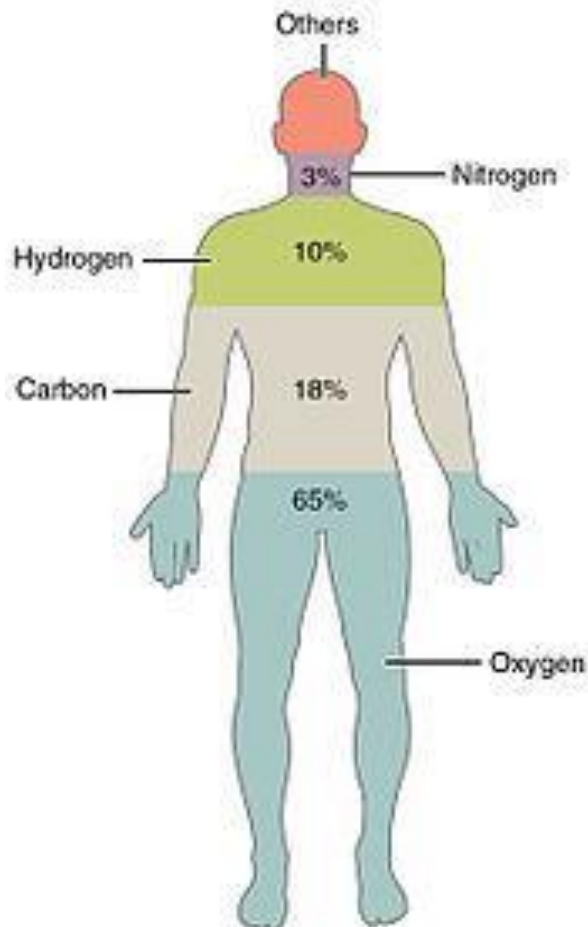
What is Chemistry?

Chemistry is a branch of natural science that deals principally with the **properties of substances**, the **changes they undergo**, and the natural **laws that describe these changes**.

Thus, **central concerns of chemistry** are -

- the structure and behaviour of atoms (elements)
- the composition and properties of compounds
- the reactions between substances with their accompanying energy exchange
- the laws that unite these phenomena

Human beings are made of chemicals?



Element	Symbol	Percentage in Body
Oxygen	O	65.0
Carbon	C	18.5
Hydrogen	H	9.5
Nitrogen	N	3.2
Calcium	Ca	1.5
Phosphorus	P	1.0
Potassium	K	0.4
Sulfur	S	0.3
Sodium	Na	0.2
Chlorine	Cl	0.2
Magnesium	Mg	0.1
Trace elements include boron (B), chromium (Cr), cobalt (Co), copper (Cu), fluorine (F), iodine (I), iron (Fe), manganese (Mn), molybdenum (Mo), selenium (Se), silicon (Si), tin (Sn), vanadium (V), and zinc (Zn).		less than 1.0

Human beings are made of chemicals?

- Almost 99% of the mass of the human body is made up of **six elements**: oxygen, carbon, hydrogen, nitrogen, calcium, and phosphorus.
 - Only about 0.85% is composed of **another five elements**: potassium, sulfur, sodium, chlorine, and magnesium.
- All are necessary to life.

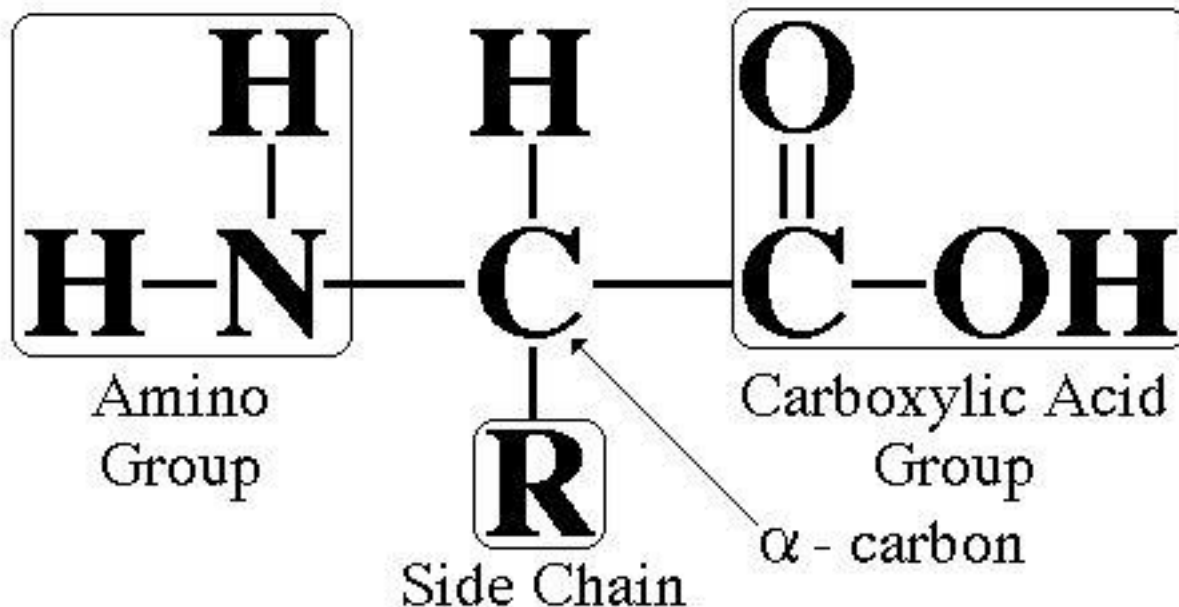
Elements and their functions in human body

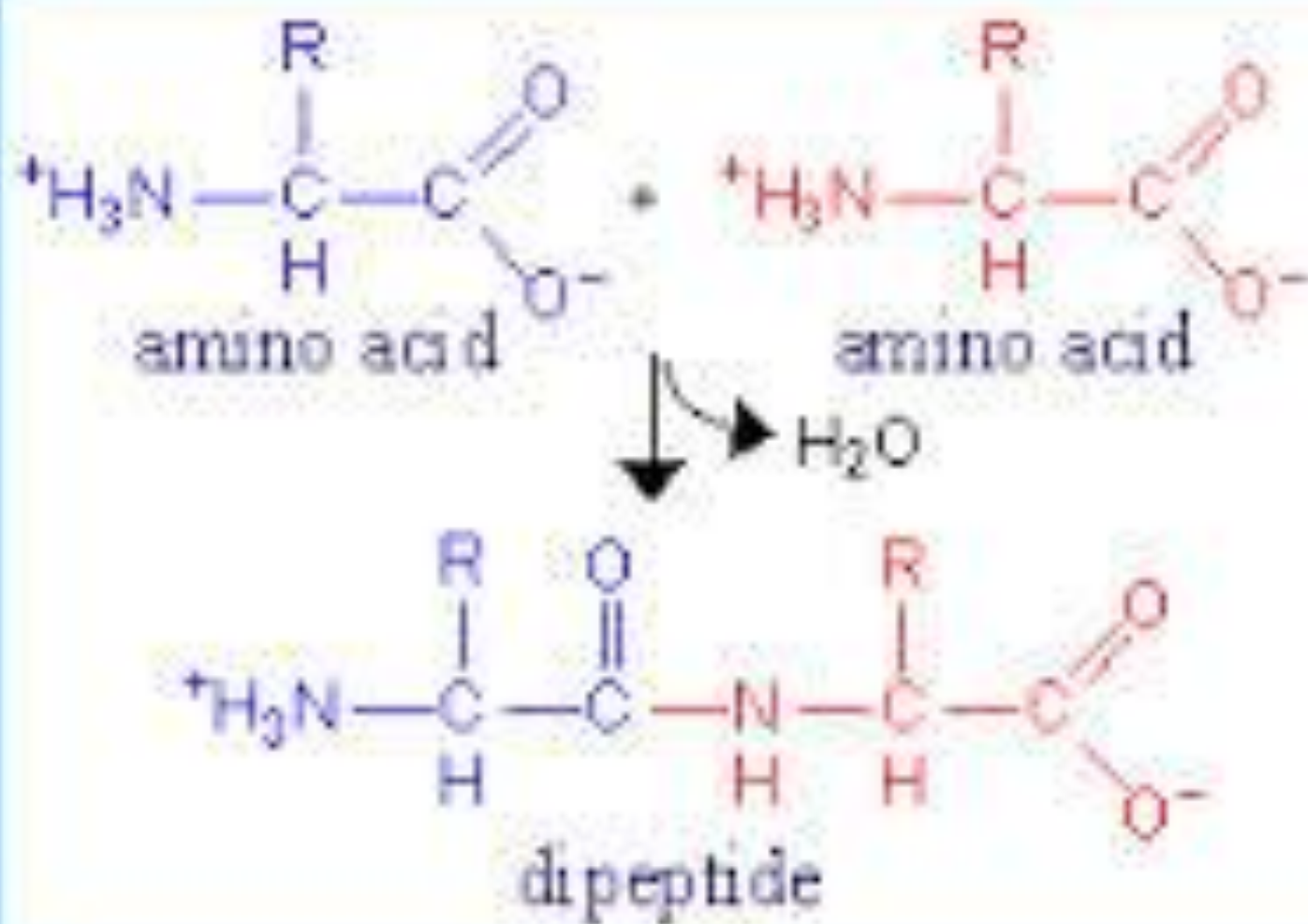
Table 2A Elements and Their Functions in the Human Body

Element	Function	Element	Function
The Big Four		The Trace Elements	
Carbon (C)	The subject of Chapters 10–19 (organic chemistry) and 20–31 (biochemistry)	Chromium (Cr)	Increases effectiveness of insulin
Hydrogen (H)		Cobalt (Co)	Part of vitamin B ₁₂
Nitrogen (N)		Copper (Cu)	Strengthens bones; assists in enzyme activity
Oxygen (O)		Fluorine (F)	Reduces the incidence of dental cavities
The Next Seven		Iodine (I)	An essential part of thyroid hormones
Calcium (Ca)	Strengthens bones and teeth; aids blood clotting	Iron (Fe)	An essential part of some proteins, such as hemoglobin, myoglobin, cytochromes, and FeS proteins
Chlorine (Cl)	Necessary for normal growth and development	Manganese (Mn)	Present in bone-forming enzymes; aids in fat and carbohydrate metabolism
Magnesium (Mg)	Helps nerve and muscle action; present in bones	Molybdenum (Mo)	Helps regulate electrical balance in body fluids
Phosphorus (P)	Present as phosphates in bone, in nucleic acids (DNA and RNA), and involved in energy storage and transfer	Zinc (Zn)	Necessary for the action of certain enzymes
Potassium (K)	Helps regulate electrical balance in body fluids; essential for nerve conduction		
Sulfur (S)	An essential component of proteins		
Sodium (Na)	Helps regulate electrical balance in body fluids		

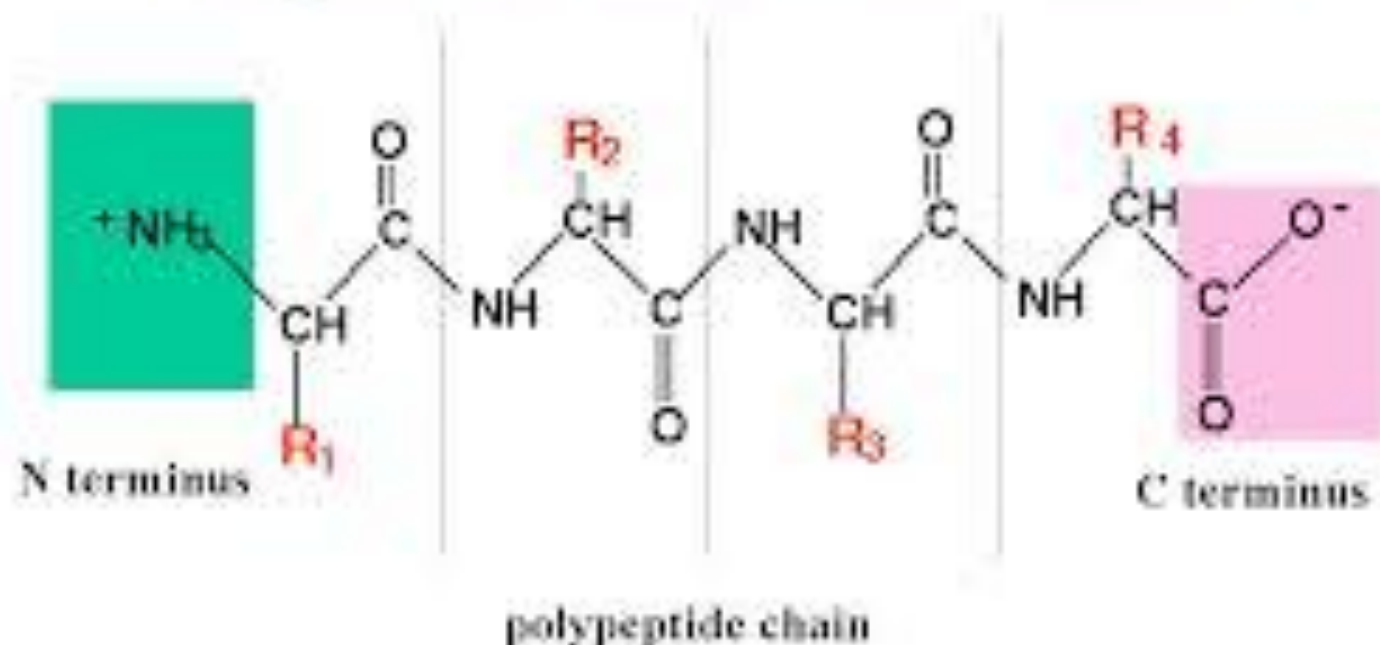
Our human body is made of amino acid

Amino Acid Structure

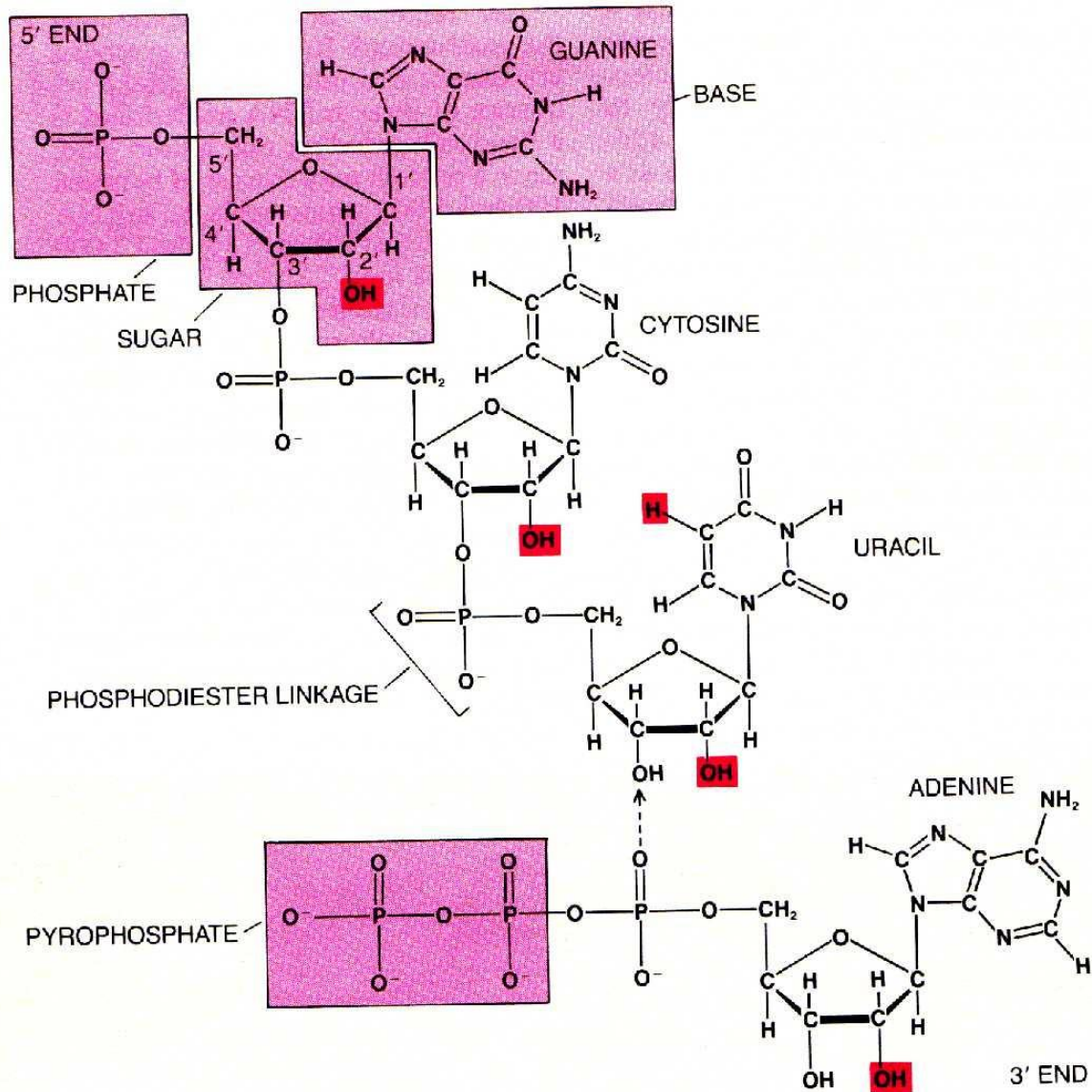




Peptide = chain of amino acids

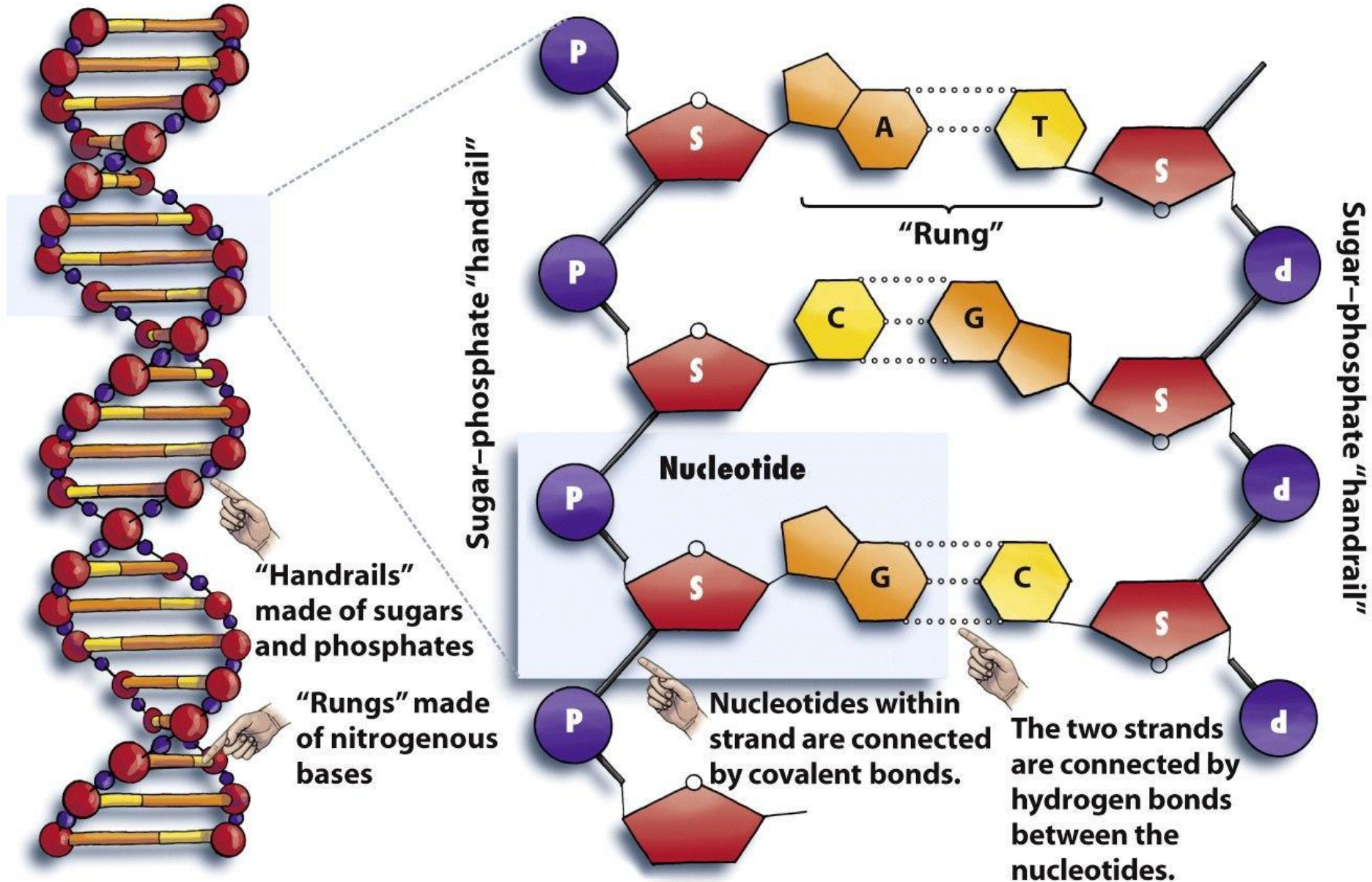


DNA



DNA double helix is made of two strands.

Each strand is a chain of of antiparallel nucleotides.



Chemistry in Everyday Life

Do we use chemicals in our daily life?

- ☐ Chemicals are everywhere and are everything. Anything you can touch or smell or see contains one or more chemicals. Many occur naturally but some are man made.
- ☐ The principles of chemistry have been used for the benefit of mankind.

Chemistry in Everyday Life

- ❑ Think of cleanliness — the materials like soaps, detergents, household bleaches, tooth pastes, etc are all chemicals.
- ❑ Sickness and diseases remind us medicines- again chemicals

Chemistry has influenced our life so much that we do not even realise that we come across chemicals at every moment

Chemistry in Everyday Life

We will discuss application of chemistry in three important areas, namely-

Medicines

Food materials

Cleansing agents

Chemistry in Everyday Life

Medicines

- Drugs are chemicals of low molecular masses.
- Drugs Interact with macromolecular targets and produce a biological response
- When the biological response is therapeutic and useful, these chemicals are called medicines

Chemotherapy:

Use of chemicals for therapeutic effect is called chemotherapy

Some common drugs

Antacids

□ An **antacid** is a substance which neutralizes stomach acidity.



Major function of antacids is to neutralize excess HCl in gastric juice

□ pH of gastric juice is about 1.5, which correspond to HCl conc. of 0.03M – a concentration strong enough to dissolve zinc metal!

- Over production of acid in the stomach causes irritation and pain. In severe cases, ulcers are developed in the stomach.
- Treatment for acidity is to use antacids, such as sodium hydrogencarbonate or a mixture of aluminium and magnesium hydroxide.
- However, excessive hydrogencarbonate can make the stomach alkaline and trigger the production of even more acid.
- Metal hydroxides are better alternatives because of being insoluble, these do not increase the pH above neutrality.

Active ingredients of some popular antacids:

Some Common Commercial Antacid Preparations	
Commercial Name	Active Ingredients
Alka-2	Calcium carbonate
Alka-Seltzer	Aspirin, sodium bicarbonate, citric acid
Bufferin	Aspirin, magnesium carbonate, aluminum glycinate
Buffered aspirin	Aspirin, magnesium carbonate, aluminum hydroxide-glycine
Milk of magnesia	Magnesium hydroxide
Rolaids	Dihydroxy aluminum sodium carbonate
Tums	Calcium carbonate

Reactions by which antacids neutralize stomach acid are as follows:

